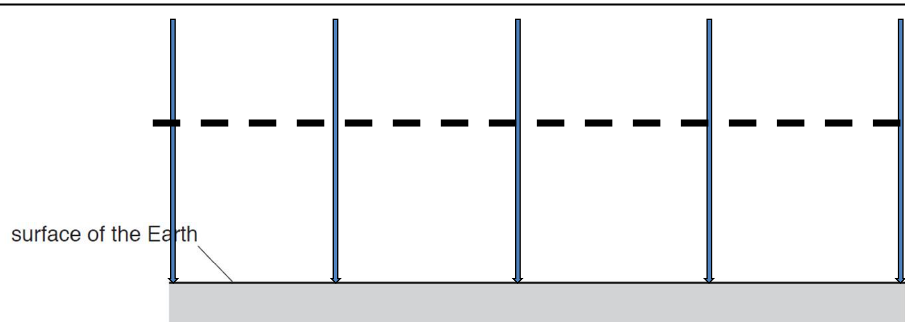
2 (a)	$g = F / m$ {i.e. definition of g is 'force per unit mass'} $GMm / r^2 = mg$ {Newton's law of gravitation = F} $GM / r^2 = g$	[1]
(b)(i)1.	6300 ± 200 km	[1]
(b)(i)2.	$g = 6.0 \pm 0.2$ N kg^{-1} force = mg = 20 000 × 6.0 = 120 000 ± 4000 N Do not accept: "9.81" as it is not reading off the graph	[1] [1]
(b)(ii)	answer to (b)(i)2 = m v^2 / r OR their g = v^2 / r $v^2 = 6.0 \times 8.2 \times 10^6$ $v = 7.0 \times 10^3$ m s^{-1}	[1] [1] [1]
(c)(i)	The work done per unit mass (by an external force) in bringing a small test mass from infinity to that point (without a change in its KE).	[1]
(c)(ii)	recognises that this is the area under the graph counting squares from <u>infinity to that point</u> total gravitational potential = $(-4.0 \pm 0.2) \times 10^7$ J kg^{-1} **OR** $\phi = -GM / r$ $= -gr$, and stating: "Earth is behaving as a point/spherical mass" [1] $= -(4.0 \pm 0.2) \times (10\,000 \times 10^3)$ [1] $= -4.0 \times 10^7$ J kg^{-1} [1]	[1] [1] [1]
(d)(i)	 <u>At least 5</u> <u>vertical</u> lines <u>equally spaced</u> (by eye) <u>Arrowheads</u> towards the surface of the Earth	[1] [1]
(d)(ii)	One dashed line <u>perpendicular</u> to the field lines drawn in (d)(i) (ecf allowed if field lines were not vertical)	[1]
(d)(iii)	for small (variation in) distances, *g* is approximately constant (and *g* varies for large distances)	[1]

	OR Fig 2.1 shows how *g* varies for large distances, while (d)(i) shows how <u>*g* is approximately constant</u> over very <u>small distance</u>. OR Height in **(d)(i)** is much smaller than the radius of Earth {Up to a height of 1000 m, *g* is approximately constant 🡐 This answer is not accepted as from Fig 2.1, there is a perceptible change in *g* over 1000 m}	